(iii) Supporting Tissues:

These tissues provide strength and flexibility to plants. They are further of two types.

(a) Collenchyma Tissues:

They are found in cortex (beneath epidermis) of young stems and in the midribs of leaves and in petals of flowers. They are made of elongated cells with unevenly thickened primary cell walls. They are flexible and function to support the organs in which they are found.

> Most parenchyma cells can develop the ability to divide and differentiate into other types of cells and they do so during the process of repairing an injury.

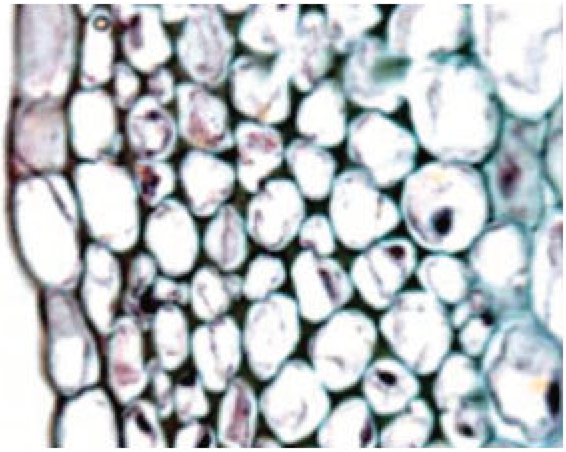


Figure 4.39 Collenchyma tissue

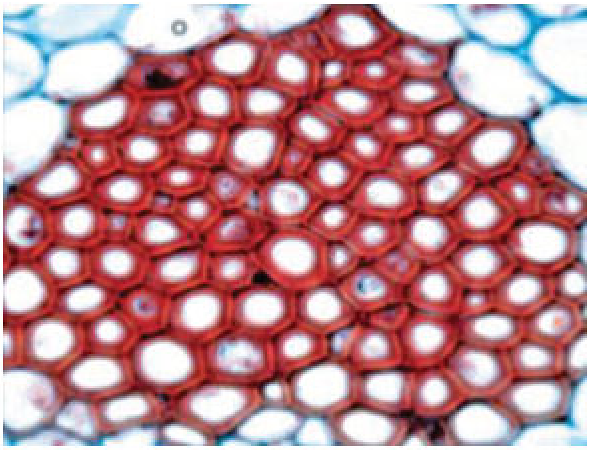


Figure 4.40 Schlerenchyma tissue

(b) Sclerenchyma Tissues :

They are composed of cells with rigid secondary cell walls. Their cell walls are hardened with lignin, which is the main chemical component of wood. Mature sclerenchyma cells cannot elongate and most of them are dead.

(B) Compound (Complex) Tissues:

A plant tissue composed of more than one type of cell is called a compound or complex tissue. Xylem and phloem tissues, found only in vascular plants, are examples of compound tissues.

(i) Xylem Tissue:

Xylem tissue is responsible for the transport of water and dissolved substances from roots to the aerial parts. Due to the presence of lignin, the secondary walls of its cells are thick and rigid. That is why xylem tissue also provides support to plant body. Two main types of cell are found in xylem tissue i.e. vessel and tracheids. **Vessels** have thick secondary cell walls. Their cells lack end walls and join together to form long tubes. **Tracheids** are made up of slender cells with overlapping ends.

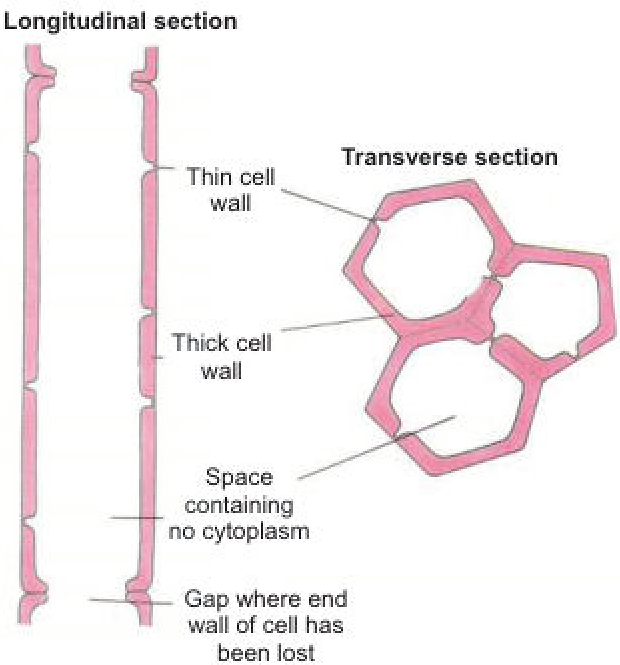


Figure 4.41 Xylem tissue

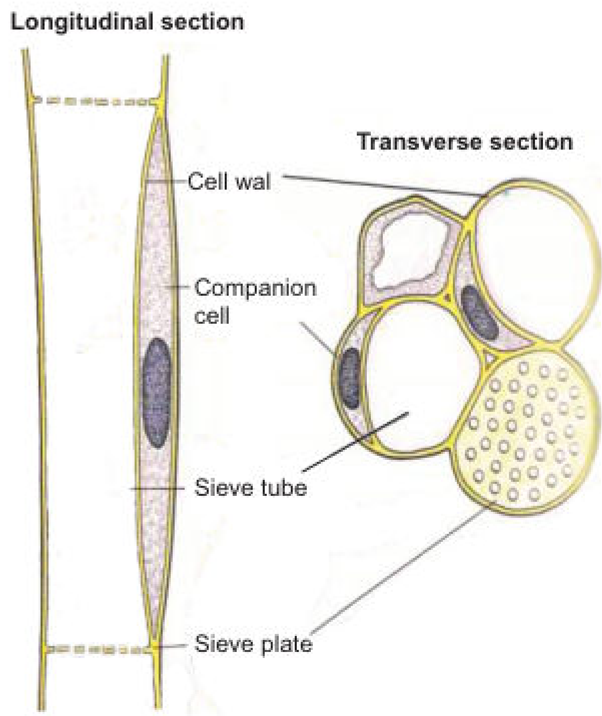


Figure 4.42 Phloem tissue

(ii) Phloem Tissue:

Phloem tissue is responsible for the conduction of dissolved organic matter (food) between different parts of plant body. Phloem tissue mainly contains sieve tube cells and companion cells. **Sieve tube cells** are long and their end walls have small pores. Many sieve tube cells join to form long sieve tubes. **Companion cells** are parenchymatous, narrow, elongated cells, and are closely associated with the sieve tube. Conduction with the sieve tube is done through the pores present on the walls of these cells. They help the sieve tubes in conduction of food materials and make proteins for sieve tube cells.

Summary

- *Zachanan Janson* is believed to be the first investigation to invent compound microscope and *Robert Hooke* improved it.
- Two parameters are important in microscopy i.e magnification and resolution.
- Another microscope is electron microscope this microscope produce higher resolution. It can be used to examine sub-cellular structures.
- A cell is the basic structural and functional unit of organisms explained by cell theory important generalization of Biology.
- There are 2 types of cells on the bases of their sub-cellular structure i.e prokaryotic and eukaryotic cell.
- The prokaryotic cell has improper nucleus i.e with out nucleus membrane while eukaryotic cell has proper nucleus surrounded by nuclear membrane.
- Cell wall is a tough, rigid, non-living, permeable outer protective layer of some cells.
- Cell- membrane is the outer most living, differentially permeable boundary of all cells.
- *S.J singer* and *G.L. Nicholson* proposed fluid mosaic model to explain the structure of cell membrane.
- Movement across cell-membrane then place through osmosis, diffusion, active transport facilitated diffusion.
- The structure present in cell called cell organelle like, Mitochondria, Golgi bodies, Endoplasmic reticulum, ribosome, lysosome, vacuoles, centrioles, plastids and nucleus.
- Cells are of variable size like bacterial cells of smallest size and egg cells of largest size.
- waste production and demand of nutrients are directly proportional to cell volume.
- Tissues are the group of similar cells may be on the basis of structure.
- In plants there are two major types of tissues i.e meristematic and permanent tissues.

Review Questions

1. Encircle the correct answer:

(i) What is responsible for the high resolution of the electron microscope?

(a) High magnification

(b) Short wavelength of the electron beam

(c) Use of heavy metals strains

(d) Very thin section

(ii) What is a function of the rough endoplasmic reticulum?

(a) Aerobic respiration (b) Intracellular digestion

(c) Synthesis of steroids (d) Synthesis of protein

(iii) Which statement about the fluid mosaic model of membrane structure is correct?

(a) The less unsaturated the fatty acid, the more fluid nature.

(b) The more unsaturated the fatty acid, the more fluid nature.

(c) Higher the temperature, less fluid nature.

(d) The lower the temperature, more fluid nature

(iv) Which process allow movement in and out of cell

I. Osmosis

II. Diffusion

III. Active transport

(a) I only (b) I and II only

(c) II and III only (d) I, II and III

(v) All are postulates of cell theory except

(a) New cell is derived from pre-existing cells.

(b) Cell does not contain the hereditary material.

(c) All living organisms are made up of one or more cells.

(d) Cell is the fundamental unit of life

(vi) Secondary wall is made up of

(a) Pectin and cellulose (b) Cellulose and protein

(c) Cellulose and lignin (d) Lignin and pectin

(vii) Select the odd one

(a) Active transport (b) Diffusion

(c) Facilitated diffusion (d) Osmosis

(viii) Trace the correct pathway of protein produce from protein factories

(a) RER → Ribosome → Golgi body → Lysosome

(b) Ribosomes → RER → Golgi body → Lysosome

(c) Golgi body → RER → Ribosome → Lysosome

(d) RER → Ribosome → Lysosome → Golgi body

(ix) Cell organelle found in animal cell and help intracellular digestion

(a) Lysosome (b) Ribosomes

(c) Mitochondria (d) Golgi apparatus

(x) Select the mismatched

(a) Plastids → Storage of chemicals

(b) Centriole → Help in cell division

(c) Ribosomes → Synthesis of steroids

(d) Mitochondria → Synthesis of ATP

2. Fill in the blanks:

(i) Microscopes are instrument designed to produce ________ visual image.

(ii) Resolution of a microscope is defined as the smallest distance between_______ points.

(iii) Magnification of a light microscope is formed by using mixture of the power of the eyepiece and the _______ lens.

(iv) Electron has a much shorter wavelength than visible light, and this allows electron microscopes to produce ___________ images.

(v) In plants, the cell wall is composed mainly of strong fibers of ___________.

(vi) Cell membrane is composed of ________ layer.

(vii) Diffusion is a _______ process, which does not require energy input.

(viii) Plant cell loses water and cytoplasm shrinks in a process called ___________.

(ix) Special type of movement of specific substances through carrier protein is ___________.

(x) The microtubules arranged in a very particular pattern to form centriole are _________ in number.

3. Define the following terms:

(i) Exocytosis (ii) Vesicles (iii) Cartilage

(iv) Nucleoplasm (v) Cyclosis (vi) Plasmolysis

(vii) Resolution (viii) Tissue (ix) Magnification

(x) Cisternae

4. Distinguish between the following in tabulated form:

(i) Prokaryotic cell and eukaryotic cell

(ii) Mitochondria and Chloroplast

(iii) Lysosome and Ribosomes

5. Write short answers of following questions:

(i) Why mitochondria is also called power house of cell?

(ii) Why iodine used to stain the onion peel?

(iii) How electron microscope is different from simple compound microscope?

(iv) Why cell membrane is semipermeable in nature?

(v) How facilitated diffusion is different from active transport?

(vi) Why cell is considered as the structural and functional unit of living things?

6. Write detailed answers of the following questions:

(i) Describe structure and function of nucleus with the help of diagram.

(ii) What is microscope? Describe types of microscopes.

(iii) Describe fluid mosaic model of cell membrane also draw the diagram.

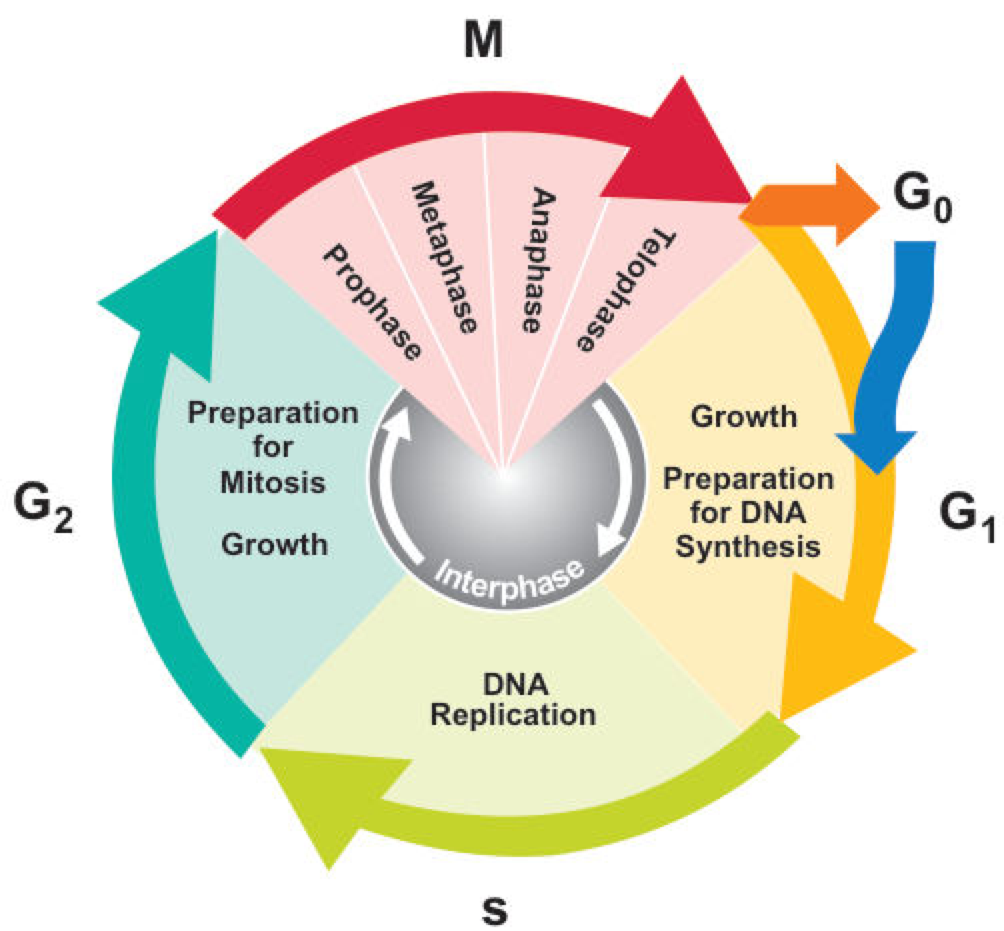
CELL CYCLE

Chapter 5

Major Concept

In this Unit you will learn:

- Chromosomes Structure and Functions
- Cell Cycle (Interphase and Division)
- Mitosis
  - Phases of Mitosis
  - Significance of Mitosis
- Necrosis and Apoptosis
- Meiosis
  - Phases of Meiosis



5.1 CHROMOSOMES

The term Chromosomes is given by German embryologist *Walter Fleming* in 1882 when he was examining the rapidly dividing cells of salamander larvae after treating with Perkin's Aniline. He observed that chromosomes colour is much darker than the rest of organelles. The term chromosomes is misnomer because its means coloured body later it was found that chromosomes are colourless bodies

Fig: 5.1 structure of chromosome

Chromosomes are thread like structure appear at the time of cell division includes found in specific numbers, made up of chromatin material in eukaryotic cell. They contain heredity units called **Genes**.

Chromosomes are made up of DNA and basic protein, Histones, appear during the cell division in the shape of rod. It has two parts arms and centromere.

The chromosomes are of different types, depending upon position of centromere. These types are:

(i) **Metacentric:** Chromosomes with equal arms.

(ii) **Sub-meta centric:** Chromosomes with un equal arms

(iii) **Acrocentric or sub-telocentric:** Rod like chromosomes with one arm very small and other very long. The centromere is subterminal.

(iv) **Telocentric:** Location of centromere at the end of chromosomes.

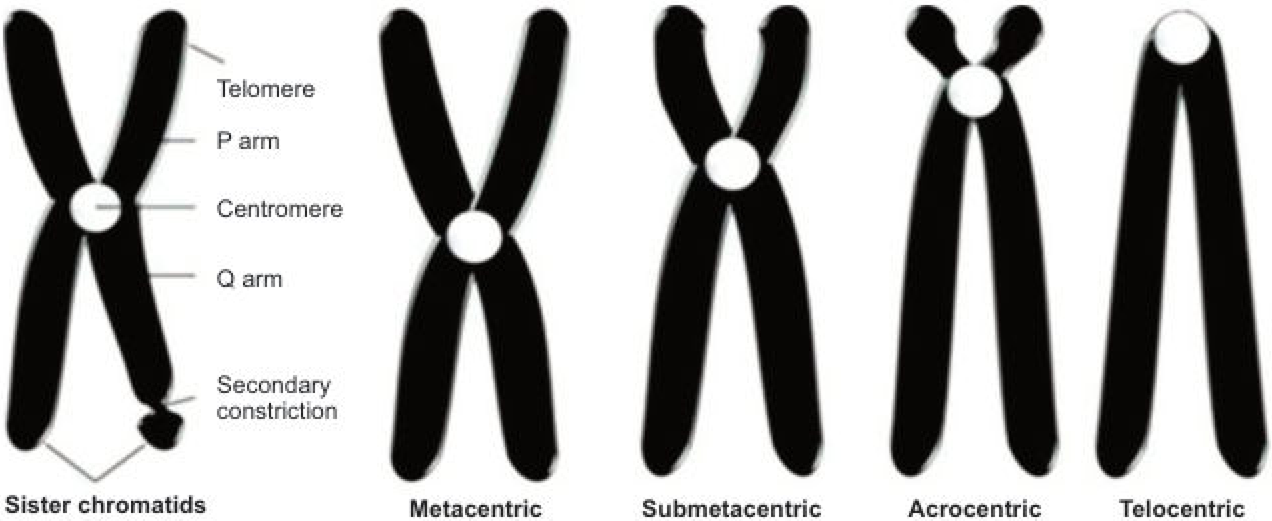


Fig: 5.2 Types of chromosomes

In the beginning of cell-division each chromosome is consist of two genetically identical copies of thread attach with each other called chromatids or sister chromatids.

Formation of chromosome:

Each chromosomes in eukaryotes are composed of chromatin fiber, which is made of nucleosomes. Chromatin fibers are packaged by proteins into a condensed structure called chromatin.

Chromatin allows the very long DNA molecules to fit into the cell nucleus. During cell division chromatin condenses further to form microscopically visible chromosomes. The structure of chromosomes varies through the cell cycle.

During cell cycle chromatin material replicate, divide and passed successfully to their daughter cells for survival of their progeny. Some time cell-division is also responsible for genetic diversity.

5.2 CELL CYCLE

The sequence of changes which occurs between one cell division and the next is called Cell Cycle." It has two phases, **Interphase**, which is the period of non-division and **M-phase**, which is a period of cell division.

The cell cycle undergoes a sequence of changes, which involve period of growth, replication of DNA followed by cell division. This sequence of changes is called **cell cycle**.

Interphase:

The period of cell cycle between two consecutive divisions is called Interphase. It is a period of growth and synthesis of DNA. During this period the cell prepares itself for the M- phase.

The Interphase is divided further into three sub-phase, G_1- phase, S-phase and G_2-phase.

G_1-(Gap one) phase: It is the period of extensive metabolic activity, in which:

Cell grows in size, specific enzymes are synthesized and DNA base units are accumulated for the DNA synthesis.

At a point in G_1, the cell may enter into a phase called G_0 (G-knot) where cell cycle stop. It remains for days, weeks or in some cases even for the life time of the organism.

S-(Synthesis) phase: During this phase, replication of DNA occurs. As a result of it chromatin material is duplicated.

G_2– (Gap two) phase: (Pre-Mitotic Phase): The following changes occur during this phase: Cell grows in size, cell organelles are replicate in numbers as well as enzyme require for cell-division also synthesized during this phase.

5.3 MITOSIS

In this type of cell division a parent cell divides into two daughter cells in a way that the number of chromosomes in the daughter cells remains the same as in the parent cell.

Although mitosis is a continuous process, but for the study point of view we can divide it into two phases; (a) Karyokinesis - nuclear division (b) Cytokinesis - cytoplasmic division.

(a) The karyokinesis can be divided further for convenience into four phases which are **Prophase**, **Metaphase**, **Anaphase** and **Telophase**. Let us study mitosis in an animal cell.

(i) Prophase:

During early prophase chromatin material condenses and become visible as thick coiled, thread like structures called **chromosomes**. Each chromosome at this stage is already double, consists of two **chromatids**. The chromatids are attached to each other at **centromere**. The nuclear membrane gradually disappears and at the same time centrosome divides to form two centrioles, each moves towards the opposite pole of the animal cell and forms the spindle fibres. The centrioles are absent in plant cells.

(ii) Metaphase:

During this phase each chromosome arranges itself on the equator of the spindle. Each chromosome is attached to separate spindle fibre by its centromere.

(iii) Anaphase:

In this phase the spindle fibre contract, centromere of a chromosome divides and the chromatides of each chromosome separates from each other and begin to move towards the other poles. In this way one set of the chromatids (each chromatid is now an independent chromosome) move towards one pole while the other set towards the other pole.

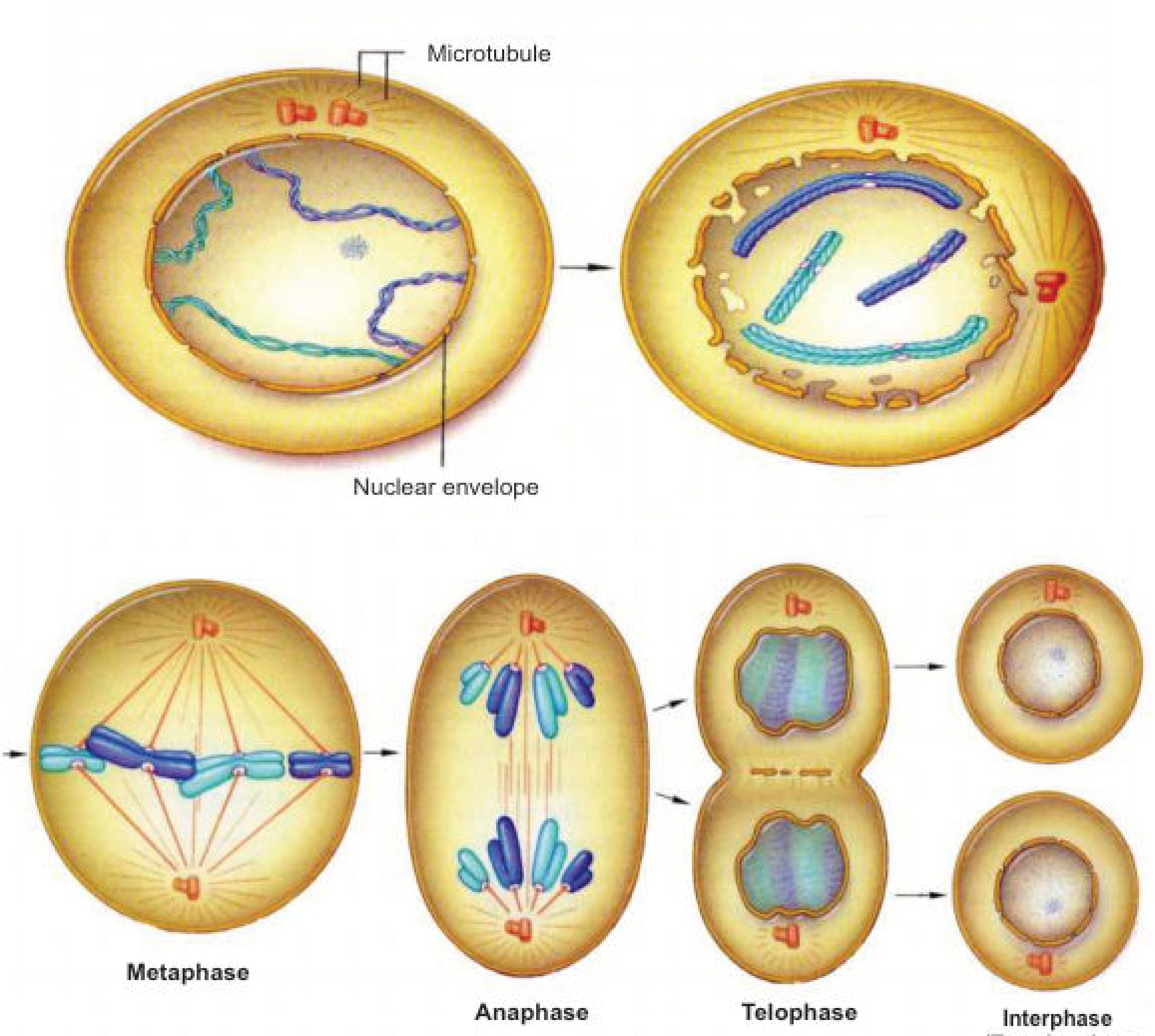


Fig: 5.3 Different stages of Mitosis

(iv) Telophase:

This is a stage when the chromatids (now called chromosomes) reach at the respective poles and their movement ceases. Each pole receives the same number of chromosomes as were present in the parent cell. The nuclear membrane is reformed around each set of chromosomes. In this way two daughter nuclei formed in each cell.

(b) Cytokinesis; soon the cytoplasm of the cell also divides and two daughter cells are formed. In animal cell cytokinesis takes place by developing a constriction. This constriction become deep to divide cytoplasm in two equal halves and two daughter cells are formed. In plant cells it occurs by developing cell plate. In this way the daughter cells become the exact copies of their parent cell.

Significance of mitosis:

Mitosis plays an important role in the life of an organism. It is responsible for development and growth of organisms by increasing exact copies of cells. With few exception all kinds of asexual reproduction and vegetative propagation take place by mitosis. The production of new somatic cells, such as blood cells depends on mitosis. The healing of wounds, repair of wear and tear within organism is also dependent upon the mitotic division.

5.4 APOPTOSIS AND NECROSIS (two ways of cell death)

Cell in an organism depends upon various extra cellular signals for its regulated and controlled activities. It means all the activities even the death of cells is programmed.

Is cell death beneficial?

Programmed cell death helps in proper control of multicellular development, which may lead to deletion of entire structure, e.g. the tail of developing human embryo, or some part an organ which is more required like tissue between developing digits.

Two ways of cell death in Multicellular organisms:

Apoptosis or Self - Destruction (Autophagy): "Programed change which lead to sequence of physiological changes in cell by which cells commit suicide collectively called **Apoptosis**".

Necrosis:

This type of cell death which is caused by external factors i.e infection, toxin and tumor i.e accidental cell death.

5.5 MEIOSIS (Reduction Division)

Meiosis is a type of cell division in which single cell divides into four daughter cells and number of chromosomes becomes half in each daughter cell.

In animal meiosis takes place in germ cell to produce gametes i.e. Sperms and Eggs whereas in plants it takes place in spore mother cells (S.M.C) to produce spores.

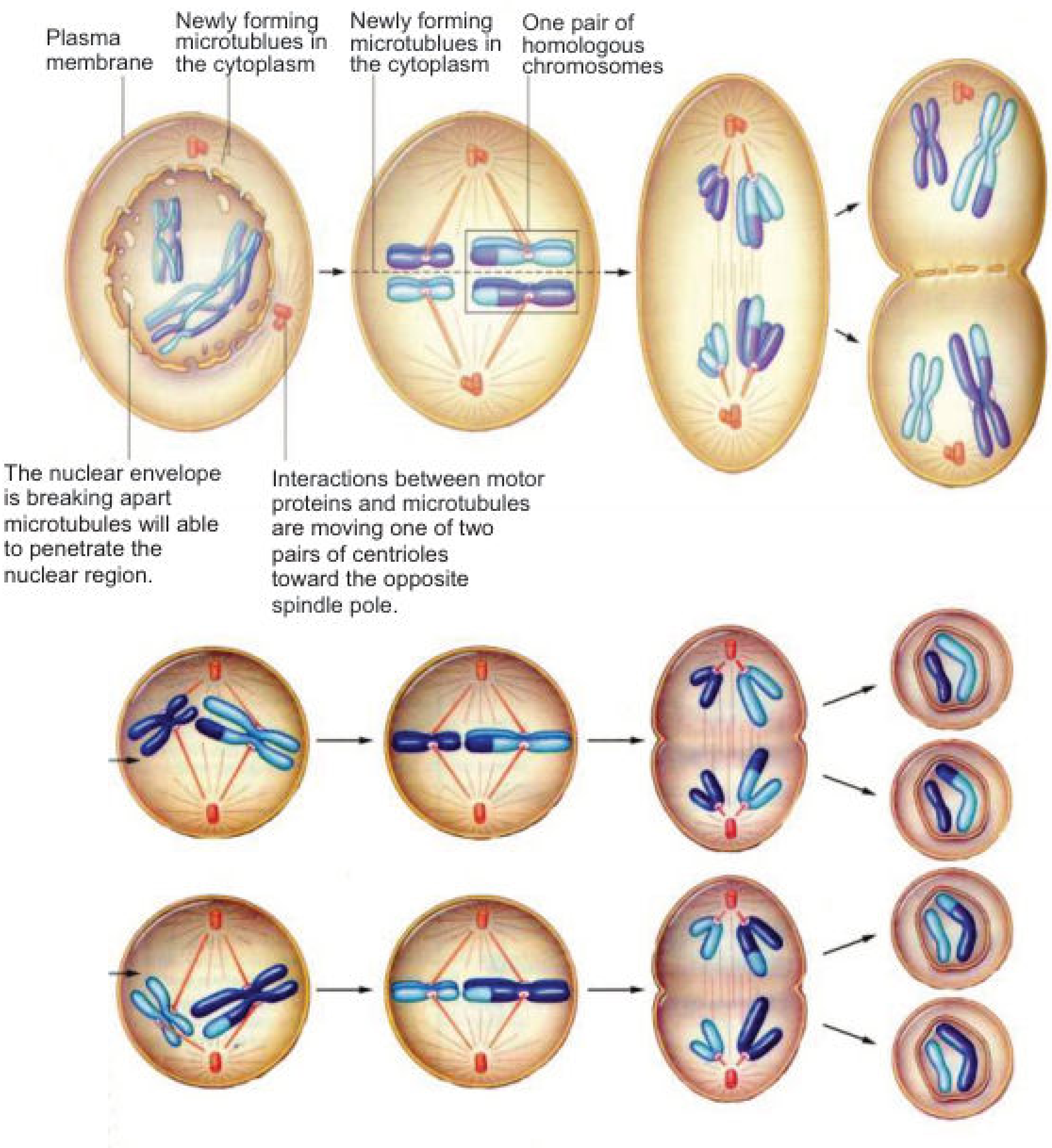


Fig: 5.4 Different stages of Meiosis

Events of Meiosis:

Meiosis is a series of two divisions, MEIOSIS I and MEIOSIS II which result in the formation of four haploid cells.

Meiosis I (First Meiotic Division)

First meiotic division is the reduction division during which the chromosomes number is reduced to half. Meiosis I consists of Prophase I, Metaphase I, Anaphase I and Telophase I.